

Known Adversarial Attacks by 2023

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Abstract — AI is one of the most popular field of technologies nowadays. Developers implement these technologies everywhere forgetting sometimes about its robustness to unobvious types of traffic. This omission can be used by attackers, who are always seeking to develop new attacks. So, the growth of AI is highly correlates with the rise of adversarial attacks. Adversarial attacks or adversarial machine learning is a technique when attackers attempt to fool ML systems with deceptive data. They can use inconspicuous, natural-looking perturbations in input data to mislead neural networks without inferring into a model directly and often without the risk to be detected. Adversarial attacks usually are divided into three primary axes — the security violation, poisoning and evasion attacks, which further can be categorized on “targeted”, “untargeted”, “whitebox” and “blackbox” types. This research examines most of the adversarial attacks are known by 2023 relating to all these categories and some others.

Keywords—adversarial attack, machine learning models, white-box, black-box, poisoning attack, evasion attack.

I. INTRODUCTION

The field of artificial intelligence has rapidly grown in popularity and has found widespread application in various industries, including computer vision. However, the robustness of these technologies to unusual types of inputs can often be overlooked. This presents an opportunity for adversaries to launch attacks and manipulate the models to produce incorrect predictions. The goal of an adversarial attack is to create such input that it appears normal but causes the model to make an error in prediction. This continuous process of development and defense has grown along with the rise of artificial intelligence, with attackers always seeking new ways to exploit weakness in machine learning models.

Attacks on ML are possible because of at least two reasons. Firstly, there is the fact which guarantees the similarity between the input spaces of default image and perturbed one, they both lay on the same space and attacked image still represents image. Secondly, adversarial attack can be analyzed in interpretable way.

Adversarial attacks usually are divided into three primary axes — the *security violation*, *poisoning* and *evasion* attacks, which further can be categorized on *targeted* and *untargeted*, *white-box* and *black-box*, *one-shot* and *iterative*, consist of *specific* or *universal* perturbations. This research examines most of the adversarial attacks known by 2023 relating to all these categories and some others.

II. PRIMARY TYPES OF ADVERSARIAL ATTACKS

In this section we introduced the main vectors of adversarial attacks and their characteristics.

A. Security violation attacks

Security violation attacks – active and ongoing research area, with new methods and technics being developed. Attacks where the attacker attempts to extract sensitive information from the machine learning model or its training data. These attacks can take many forms, including:

Model inversion attack: an attacker attempts to reconstruct the training data from the model’s parameters.

Membership inference attack: an attacker attempts to distinguish between examples are used in the training set and not, using some auxiliary information, such as model’s performance on a specific input or output.

Model stealing or extraction attack: an attacker attempts to obtain a copy of the model’s parameters or its architecture by making queries to the model with the goal to replicate the model, modify it or cause the model to malfunction.

The consequences of these attacks are serious, such as compromising the privacy of individuals or organizations or allowing unauthorized access to restricted information. Recent researches, such as differential privacy, secure multi-party computational, adding noise to the model’s output or removing identifying information, model encryption, limiting the number of queries to the model and etc., have been proposed as the methods to improve the security level and to defend from this type of attacks.

B. Poisoning attacks

Poisoning attacks –attacks refer to a type of attack on ML models where the attacker manipulates the training data in order to cause the model to make incorrect predictions on a specific subset of inputs. The main goal of a poisoning attack is to corrupt the model’s decision-making process by introducing malicious data into the training set, with the aim to cause the model to behave incorrectly.

For example, an attacker may try to add a few malicious examples to a dataset used for training a spam filter, in order to cause this filter to classify some illegitimate emails as non-spam, or vice-versa.

There are several methods that can be used to perform poisoning attacks, including:

- Label flipping: the attacker changes the labels of a subset of the training data.
- Backdoor attacks: where the attacker adds a small number of examples to train set that are designed to cause the model to make a specific mistake when it encounters certain input.
- Data poisoning in resources with open datasets: the attacker adds subset of examples which can misclassified the model, and it is very difficult to detect because of enormous dataset size.

Poisoning attacks can be difficult to detect, as a model may continue to perform well on the majority of inputs but will make mistakes on a specific number of inputs. Additionally, the attacker can choose to target a specific class of inputs, making the attack even more difficult to detect. To mitigate these attacks, recent research has proposed techniques as validate training data before using it to train the model, for example, data cleaning, outlier or anomaly detection, data integrity checks and other.

C. Evasion attacks

Evasion attacks — this is the type of attacks, where data are modified to evade detection or to be classified as legitimate. Evasion attacks are aimed to testing dataset and don't affect on the training set. These attacks are dominant in modern world and further can be classified according to these characteristics:

Targeted attack – the type of attack where an attacker introduces perturbations which make the model classify this example in some specific label.

Untargeted attack – the type of attack where all things an attacker cares is making the model to make incorrect labeling, it does not matter what specific label, just any other than the right one.

One-shot attack – the type of attack where the attacker needs to do only one calculation to make the model be misclassified with this perturbed example.

Iterative attack – the type of attack where the attacker needs to do a lot calculation one by one to generate adversarial example. At each iteration algorithm finds the direction of the smallest perturbation that causes the model's prediction to change, and adds a small magnitude of this direction to the input. The process continues until the model's prediction has changed to the desired target class. This attack takes more time than one-shot attack, but usually is more effective.

Universal perturbation – the type of attack where the attacker adds the same universal perturbation to all the inputs.

Specific perturbation – the type of attack where the attacker adds unique perturbations to each of the inputs.

Usually, adversarial attack is defined in such way: let $x_0 \in \mathbb{R}^d$ is a default image with a label y_0 . Attacker adds a perturbation $r \in \mathbb{R}^d$, changing the default image to $x^* = x_0 + r$ with a label y^* and $y_0 \neq y^*$.

III. WHITE-BOX ATTACKS

White-Box attacks refer to adversarial attacks where the attacker has full information of the target model, e.g., its architecture, parameters, and gradients. In these attacks, the attacker can leverage this information to craft adversarial examples that are specifically designed to fool the target model. White-box attacks provide a stronger scenario, as the attacker has a complete understanding of the model's behavior and can optimize their attacks accordingly.

A. Fast Gradient Sign Method

The first researcher who discovered the foolish “model misclassification” by changing even a few pixels in image was Christian Szegedy et al. in 2014 [1]. After that moment many researchers successfully proposed efficient gradient-based attacks, first of which was Fast Gradient Sign Method (FGSM) described by Goodfellow et al. [2].

FGSM – is quite simple targeted attack which uses the gradient sign at each pixel to determine the gradient's maximum direction for changing the corresponding pixel value and adds the perturbation δ to the input image x in such way:

$$\delta = \epsilon * \text{sign}(\nabla_x J(x, y, \theta)), \quad (1)$$

where δ represents the perturbation, ϵ means the constant which regulates the “number” of perturbations, $J(x, y, \theta)$ is the loss function, where x is the default or “clear” image, y is the corresponding to x label and θ denotes model's parameters.

And adversarial image x^* :

$$x^* = x + \delta \quad (2)$$

B. Iterative Attack

The Iterative Attack, also known as I-FGSM or Iter-LL, is a variation of the FGSM attack that involves repeating the FGSM or Step-LL process multiple times, each time with a step size $\alpha \geq \epsilon/k$ and projected onto the l_∞ ball of norm ϵ around x [3].

C. Randomized Fast Gradient Sign Method

Randomized Fast Gradient Sign Method (RAND+FGSM) proposed in [3] is one more version of FGSM which can be effective against models trained on adversarial examples. It is the same as FGSM but using random perturbation before using FGSM directly.

D. One Step Least Likely attack

One Step Least Likely attack (Step-LL) – another variant of FGSM attack propounded by Kurakin et al. [4] which differs from FGSM only in the attempt to shift a label to the least-likely class and adversarial image looks like:

$$x^* = x - \delta \quad (3)$$

One-Step method typically requires the calculation of only a single gradient and is often based on determining the optimal perturbation through a linear approximation of the cost or model.

E. Basic Iterative method

Basic Iterative method, or BIM, – is an expansion of FGSM that performs the attack repeatedly with a slight step size:

$$x_0^* = x, \quad x_{N+1}^* = \text{Clip}_{x, \epsilon} \{x_N^* + \alpha \text{sign}(\nabla_x J(x_N^*, y, \theta))\} \quad (4)$$

Where $\text{Clip}_{x, \epsilon}(A)$ refers element-wise clipping A , with referring $A_{i,j}$ clipped to the limited range $[x_{i,j} - \epsilon, x_{i,j} + \epsilon]$ and α is the value of the number of changes on each pixel on each step [5]. Iterative methods are known for more harmful influence on model's accuracy than one-step ones.

F. Jacobian-based Saliency Map attack

Jacobian-based Saliency Map attack (JSMA) – the white-box targeted attack method first time suggested by researches from [5]. It utilizes the gradient loss with each class label with respect to each input's component, in other words, the Jacobian matrix, to determine the sensitivity direction. This allows for the calculation of the forward derivative of the machine learning model to the input example x , which in turn provides an understanding of the extent of impact each pixel point has on the model's classification:

$$\nabla F(x) = \frac{dF(x)}{dx} = \left[\frac{dF_i}{dx_i} \right]_{i=1}^n, \quad (5)$$

To quantify the effect of altering pixel values on the target classifier, JSMA suggests building an adversarial significance graph using the Jacobian matrix, as shown below:

$$s^+(x, c)[i] = \begin{cases} 0, & \frac{\partial F_c(x)}{\partial x_i} < 0 \text{ or } \sum_{c' \neq c} \frac{\partial F_{c'}(x)}{\partial x_i} > 0 \\ \frac{\partial F_c(x)}{\partial x_i} \left| \sum_{c' \neq c} \frac{\partial F_{c'}(x)}{\partial x_i} \right|, & \text{otherwise} \end{cases} \quad (6)$$

where x is the input of the model, c denotes a target class of x , i represents the input feature and s^+ measures how much x_i positively correlates with c , while also negatively correlates with all other classes $c' \neq c$. The saliency resets to zero if either condition is not satisfied.

The above equation can help to determine which changes in pixel positions can have the most significant impact on target classification. The map rates each input feature and when the value is higher, the target classifier is more vulnerable to perturbations in that feature. The attacker can exploit this vulnerability by targeting an adversarial class that does not match the true class label of the input example. By increasing a few high-saliency pixels, the prediction confidence of the adversarial class in the perturbed image increases, potentially resulting in misclassification. It is also possible to perform a decreasing operation using another saliency map that differs from the original by inverting low-saliency inequalities [6], i.e.:

$$s^-(x, c) = 0 \text{ if } \frac{\partial F_c(x)}{\partial x_i} > 0 \text{ or } \sum_{c' \neq c} \frac{\partial F_{c'}(x)}{\partial x_i} < 0 \quad (7)$$

The main contrast between JSMA and FGSM lies in the reduction of perturbations in JSMA, resulting in adversarial examples that are less noticeable. However, this comes at the cost of increased computational cost.

G. Carlini-Wagner

Carlini-Wagner (CW) – is an optimization-based targeted attack so it is solving an optimization problem. However, C&W attack is computationally expensive, as it requires solving an optimization problem for each input. This method is based on constraining l_0 , l_2 and l_∞ norm, defining general perturbations in such way:

$$\text{minimize } D(x, x + \delta) + c \cdot f(x + \delta), \quad (8)$$

where $x + \delta \in [0, 1]^n$, D denotes constraint paradigms e.g., l_0 , l_2 and l_∞ . l_0 norm corresponds to the number of original pixels that have been altered in the generation process, l_2 remains the degree of perturbation among overall pixels, l_∞ constraint the maximum allow perturbation per pixel, c denotes the hyperparameter, and f adopts a variety of objective functions.

This method proposed by Carlini and Wagner [7] against the defensive distillation method introduced by Papernot et al. [8] which proposed to use transfer knowledge of the complex neural network to the simple version of it, that is why the attacker cannot affect to the original model.

H. DeepFool

DeepFool – is a non-targeted attack, uses iterative approach and it does not required any information about the gradients of the model. This fact makes DeepFool attack be fast and computationally efficient compared to other optimization-based attack methods. Additionally, generated

adversarial examples are usually extremely visually similar to the original input.

Moosavi-Dezfooli et al. presented this method in their work [9], which uses gradient iteration to create adversarial perturbations. DeepFool involves obtaining perturbations through calculations and then continuously adjusting the pixels of normal examples iteratively until adversarial examples are obtained through crossing the boundary of decision making. DeepFool evaluates the robustness of input examples by determining the smallest possible distance between the decision boundary of normal examples and adversarial examples. In comparison to the FGSM single-step attack, DeepFool is able to produce more precise perturbations in a quicker manner, however it is limited to only achieving non-targeted attacks.

I. AdvGan

AdvGan – is a targeted attack and can either be executed as a white-box or a black-box attack, in which the attacker creates adversarial examples utilizing generative adversarial networks (GANs). GANs learn and imitate the distribution of original instances. At the first step the attacker creates the adversarial perturbations with generator, then inputs changed image to the model and to the discriminator simultaneously. The main goals are to guide generator produce the best perturbations and to train discriminator to identify these perturbed examples efficiently.

This kind of attack usually is aimed to foolish deep fake detectors or accelerate adversarial training as the defense [10].

IV. BLACK-BOX ATTACKS

Black-box attacks, on the other hand, refer to adversarial attacks where the attacker has limited or no knowledge of the target model. In these attacks, the attacker only has access to the model's input and output and must rely on trial-and-error or other methods to generate adversarial examples. Black-box attacks are a weaker attack scenario, as the attacker does not have the same level of information about the target model as a white-box attack. However, black-box attacks are more realistic, as in many real-world scenarios, the attacker may have limited knowledge of the target model.

A. Monte Carlo Fast Gradient Sign Method

Monte Carlo Fast Gradient Sign Method (MC-FGSM) – the method where the attacker at first approximates the gradient of the input variable via Monte Carlo sampling technique, and then the gradient-based method is applied to generate adversarial attacks. A strict mathematical proof has shown the gradient estimation is unbiased and the time complexity is $O(1)$, while the existing methods are $O(N)$. For more information it is worth to look at the original research [11].

B. UPSET and ANGRI

UPSET and ANGRI – these two black-box attacks were proposed by Sarkar et al. [12], the first one creates general perturbations for a specific target class and the other one produces image specific perturbations. UPSET part generates an adversarial perturbation R through the residual generation network. Given a chosen target category t , the adversarial perturbation is determined as $r_t = R(t)$. The overall formula for generating adversarial examples is as follows:

$$x' = U(x, t) = \max(\min(s * R(t) + x, 1) - 1), \quad (8)$$

where U represents the UPSET network, and s is the size which adjusts the perturbation r_c . The loss function of the

UPSET network consists of two parts, as follows:

$$L(x, x', t) = L_c(x', t) + L_f(x', t) = \sum_{i=1}^m \log(C_i(x')[t]) + w\|x' - x\|_2^2 \quad (9)$$

where L_c punishes for the failure to generate the target attack class, and L_f used to ensure similarity between adversarial and original examples. Compared with UPSET, ANGRI concatenates A_t and A_x to obtain A_c , and then produces adversarial examples x' . The advantage of ANGRI is that it is able to generate better adversarial images, even if the classifier was trained using noisy images. This is because ANGRI can still access the input images and produce more accurate and effective adversarial examples.

C. HopSkipJump

HopSkipJump – method which was introduced by Chen et al. in their paper "HopSkipJumpAttack: A Query-Efficient Decision-Based Attack," [13] and assumed both as targeted as non-targeted work. Each iteration of the attack consists of three steps: firstly, it iterates from the last iteration, which is pushed towards the boundary via binary search, secondly it estimates the gradient direction with special equation, and at the end the updating step size along gradient direction is initialized as another equation and is decreased via geometric progression until perturbation becomes successful. For more information it is worth to look at the original research [13].

V. ATTACKS IN PHYSICAL WORLD

Adversarial attacks in general can be conducted in digital and in physical sphere. Above section describes different examples of the first one which are being produced in laboratory environment. This small section concentrates on creating adversarial examples applied to the physical world by implementing perturbations with object contrast, brightness, occlusion, etc.

A. Adversarial patch attack

Adversarial patch attack – the method which involves affixing a small printed patch on the target object to manipulate its appearance which is consistent with physical deployable. However, this technique is limited to flat surfaces and a specific viewpoint, making it unsuitable for non-planar objects and limiting its real-world applications. These constraints hinder the effectiveness of the adversarial patch method. These limitations hinder the practical usefulness of adversarial patches. The researches from [14] demonstrate several examples of such attacks where stop signs with applied patches on it causing misclassification behavior on traffic sign classifier.

B. Adversarial camouflage texture attack

Adversarial camouflage texture attack – the attacks that aim to alter the appearance of 3D objects by utilizing an optimized adversarial texture. This texture can be applied to the surface of the target object, such as a pattern on clothing or graffiti on a vehicle. The perturbation is effective for non-planar objects, making it suitable for use in physical attacks. There are two main techniques for optimizing the texture. One involves optimizing a 2D image to serve as the texture, while the other involves optimizing a 3D texture with the aid of a physical renderer. The research [15] describes the attack

where people wearing a t-shirt with a special constructed pattern are not being detected as human objects by object-detector model.

C. Optical attacks

Optical attacks – in contrast to adversarial patch and camouflage texture attacks, optical attacks aim to manipulate with light or laser to carry out physical attacks. The attacker uses a lighting device, such as a projector or laser emitter, to direct towards the target object and alter its appearance, thereby attacking the target model. Optical attacks are highly controllable and stealthy, but their use is limited by their sensitivity to environmental light, as strong light can render the adversarial light ineffective. Thus, the practical utilization of optical attacks is restricted. Examples of such attacks introduced by Gnanasambandam et al. [16] who implements an adversarial attack through an artificially designed lighting distribution.

VI. CONCLUSION

In conclusion, adversarial attacks on computer vision models pose a significant challenge to the security and reliability of artificial intelligence systems. Adversarial attacks exploit the vulnerabilities in these models to produce misleading outputs that can have severe consequences, particularly in safety-critical applications such as self-driving cars, trains or planes. Therefore, it is important to develop techniques to make machine learning models more robust against adversarial attacks.

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